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Assessment of Attachment in Differentially Reared Infant Monkeys (*Macaca radiata*): Response to Separation and a Novel Environment

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To assess the developmental consequences of rearing environments for 12 infant bonnet macaques (*Macaca radiata*), infants from 2 environments, either low foraging demand (LFD) or variable foraging demand (VFD) for their mothers, were observed under 2 test conditions after the completion of the differential rearing treatment. One of the test conditions involved introduction of the rearing cohorts to a novel room; this was done in two series of four 1-hr sessions, a series before and a series after completion of the 2nd test condition. The 2nd test condition involved a 20.5-hr removal of each mother from the rearing cohort; this was repeated once a week for 3 weeks. The results of the novel room tests effectively differentiated rearing treatments in this study and suggested treatment group differences in attachment security. In contrast, the maternal separations failed to differentiate rearing treatments.

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A major aspect of this laboratory's work on the development of infant bonnet macaques (*Macaca radiata*) involves longitudinal assessment of patterns of social and affective behavior that result from manipulation of the early rearing environment. In initial studies on this topic (Plimpton & Rosenblum, 1983; Rosenblum & Sunderland, 1982), it was found that the nature of the foraging demand placed on the mother can have dramatic consequences for the ongoing affective and social behavior of the infant, as well as the infant's subsequent response to separation from mother. The response to maternal separation has been a particular focus of interest in this laboratory (Rosenblum, 1984; Rosenblum & Kaufman, 1968; Rosenblum & Plimpton, 1981) and others (Field & Reite, 1985; McKinney, 1986; Suomi, Mineka, & Harlow, 1983) for a number of years. Reactions to separation, although now known to be influenced by the conditions of separation (Caine & Reite, 1981; Coe, Wiener, & Levine, 1984; Hinde, Leighton-Shapiro, & McGinnis, 1978; Jordan, Hennessy, Gonzalez, & Levine, 1985; Reite & Snyder, 1982; Seay & Harlow, 1965; Suomi, Harlow, & Domek, 1970), have generally been viewed as a significant reflection of the infant's relationship with its mother. The initial foraging-demand study demonstrated that infants that had been maternally reared in an environment of constant foraging demand were much less able to cope with separation than were infants raised in an environment of low foraging demand (Plimpton & Rosenblum, 1983).

Longitudinal assessment also plays a key role in the study of the development of human infants. One important perspective in the human work has focused on the security of the attachment of infant to mother. In contrast to the emphasis on separation in the nonhuman primate work, Ainsworth, Bell, and Stayton (1971) suggested that a human infant's response to separation is not itself a reliable index of the security of attachment. Instead, Ainsworth et al. concluded that it is the response to the mother during their reunion after the separation that appears to provide the more relevant assessment of the attachment relationship.

Two factors render it difficult to evaluate dyadic separation as a measure of developmental outcome by comparing the findings of Ainsworth et al. (1971) to the nonhuman primate findings from this and other laboratories. First, the difference in method was confounded by a species difference; species used must certainly be viewed as a constraint on the generalizability of experimental findings. Second, the length of separation in the studies in which researchers have used the basic paradigm (Ainsworth et al., 1971; Ainsworth & Witting, 1969) was very brief (3–8 min), in comparison with the length of separation used in this laboratory (2–4 weeks) and most other primate separation studies (but see Coe, Wiener, Rosenberg, & Levine, 1985; Jordan, Coe, Patterson, & Levine, 1984).

In the previously noted study from this laboratory in which infants reared in a low-demand environment were compared with infants reared in a high-demand environment, despite differences during separation, no group differences during 2 weeks of observations after reunion were obtained. This failure to differentiate groups during reunion may have been due, at least in part, to the strong and dramatic retrieval response of the monkey mothers, which may have obscured any differences between infants. However, in keeping with the views of Ainsworth and Wittig (1969) and Bowlby (1973) that the ability of the infant to use its mother as a secure base from which to explore the environment serves as an important measure of attachment security, we have developed a test of developmental outcome based on the release of dyads into a novel environment. Using this test, we have found that bonnet macaque infants reared in a setting in which the foraging demand on mothers varied in difficulty were subsequently more reluctant to separate from mother to explore a novel environ-

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ment, and consequently were viewed as less securely attached, than were infants reared under conditions in which the foraging demands on the mothers were consistently easy.

The purpose of this study was to assess the effectiveness of the nonhuman primate separation paradigm in differentiating bonnet macaque rearing groups that previously and independently had been distinguished by novel environment tests, which had suggested differences in infant attachment security. With a number of brief, 1-day separations, we were also able to evaluate the effects of repeated dyadic separations in bonnet macaques and to make a determination of the effects of rearing conditions on the course of response to repeated separation.

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Method

Design

The subjects had participated in the previously reported studies, which used novel room testing (Andrews & Rosenblum, 1988, 1991). All subjects lived in cohorts of 3 mother–infant dyads. There were two cohorts in each of two treatment groups. The experimental rearing conditions for both treatment groups began when the infants were a mean age of 11 weeks. For the variable foraging demand (VFD) group, foraging demand on mothers varied during early infant rearing between low (food easy to obtain) and high (food difficult to obtain) in 2-week blocks for a total of 14 weeks, which began and ended with 2 weeks of low demand. In the high-demand blocks, the monkeys were required to search for their food, which was buried under 12 cm of clean woodchip bedding; the amount of buried food was only 10%–20% in excess of daily consumption. In the low-demand blocks, an amount of food at least 50% in excess of daily consumption was readily available without any digging required. In contrast, for the low foraging demand (LFD) group, foraging demand remained as in the low-demand blocks for the entire 14 weeks. Adequate food was always available for both groups, but the groups differed in the amount of time and effort required for the mothers to obtain their daily food ration. Careful health monitoring indicated normal rates of weight gain for infants in both treatment groups. The testing reported here began immediately after the completion of this differential rearing period. Throughout this study all cohorts had ad libitum access to food and water.

On each of 4 consecutive days during the 1st week after the 14-week rearing period, each cohort was given a 1-hr session in a novel environment. Over the next 3 weeks, the mother of each infant was removed from the group for 20.5 hr once each week; mothers of infants within a cohort were removed on different days. After the three separations each cohort was given four 1-hr sessions in the same novel environment during the 5th week. Findings for the initial response to the novel environment have been reported previously (Andrews & Rosenblum, 1988, 1991).

Subjects

The 6 bonnet macaque (*Macaca radiata*) dyads in each treatment group comprised two cohorts of 3 dyads each; the infants in the two cohorts within a group were born approximately 1 year apart. In each of 2 consecutive years, two cohorts, one from each treatment group, were matched for infant age and tested concurrently. There were 1 male and 5 female infants reared in each treatment condition. The mean infant age at the time of the first novel environment session was 25.3 weeks for the VFD group and 25.1 weeks for the LFD group.

Apparatus

Living area. Each cohort was housed in an indoor pen (1.8 m wide × 2.2 m high × 1.3 m deep), which was illuminated between 0700 and 1900 hr daily. Opaque walls and a one-way glass window visually isolated each pen from the outer environment. Food and water were available ad libitum.

Novel environment. The novel environment was a tile-walled pen that was more than three times the size of the home pen (16.3 m^3 vs. 5.2 m^3). Two one-way glass windows at the front of the pen permitted observations. Within the pen there were two levels of aluminum shelving as well as a number of objects that the infants had not encountered before they were introduced into the novel environment; these included colorful climbing cables, a metal feeding box (31 × 38 × 15 cm) suspended from the ceiling, and a large wooden puzzle box (55 × 55 × 25 cm) that contained an inner drum that could be rotated by reaching through a small hole on one side of the box.

Procedure

Novel environment. All dyads of a cohort were released simultaneously into the novel pen for a period of 1 hr on 4 consecutive test days of a single week. The first set of such tests were instituted in the week before the current separation assessments, and the second set of novel pen tests took place in the week after the last separation. Order of testing LFD and VFD cohorts was counterbalanced across days.

Dyadic separations. A single mother was removed from its cohort at approximately 1245 hr on one day and returned at approximately 0915 hr on the following day. The removed mother was taken to another room where it could not be seen or heard by its infant. Each mother was removed once a week for 3 consecutive weeks. Mothers in a cohort were removed on different days; the order of removal was counterbalanced across weeks.

Behavioral Observations

General. All observations consisted of a series of 5-min focal animal samples. All observations were recorded on Toshiba T-1000 laptop computers, with software that timed and prompted the observer for the various aspects of each observation and stored the recorded data for subsequent hard-copy and transfer to a desktop computer for data analysis. For focal observations on both mothers and infants, the absolute frequencies of the following behaviors were recorded: rejection or punishment of infant by mother, mother leave from dyadic contact, infant leave from dyadic contact, mother approach to dyadic contact, and infant approach to dyadic contact. For focal observations on mothers only, the frequencies of the following behaviors were recorded in addition to the dyadic behaviors: submit to another adult, dominate another adult, and distress (including vocalization, crouching, and fear grimacing). For infant focal observations only, the frequencies of social play and nonsocial play were recorded in addition to the dyadic behaviors. At 30-s intervals throughout all 5-min focal observations, the observer recorded whether each of the following behaviors was occurring: dyadic grooming, passive dyadic contact, dyadic proximity (<0.3 m apart), dyadic separation (>0.3 m), groom with nondyadic partner, passive contact with

nondyadic partner, eat, forage, and explore inanimate object. In addition, during infant focal observations the occurrence of agitation (vocalization, repetitive or stereotypic locomotion, or self-orality) or depression (hunched posture, eye closing, or head drop) was recorded every 10 s. For sustained behaviors, such as social play, the end of each occurrence (bout) was defined by a 3-s break in the behavior. Each of these behavioral categories followed standardized definitions and have been used in this laboratory with excellent interobserver reliability for over 2 decades (Plimpton, Swartz, & Rosenblum, 1981; Rosenblum, Clark, & Kaufman, 1964).

Novel room. During sessions in the novel pen, each animal in a cohort was the focal subject for one 5-min focal sample in the first half-hour and one focal sample in the second half-hour. The order in which the subjects were observed was counterbalanced across test days.

Dyadic separations. Because separations involved removal of the mother, all observations were made in the home pen; the mothers were not observed while separated. Separations were performed on one LFD and one VFD dyad concurrently. Only the animals of the two dyads currently scheduled for separation were the focal subjects during an observation period with observations alternating between the LFD and VFD cohorts. Observations were made at six time points: immediately before separation (*preseparation*), immediately after separation (*postseparation*), 3 hr after separation (*late separation*), immediately before reunion (*prereunion*), immediately after reunion (*postreunion*), and 6 hr after reunion (*late reunion*). For preseparation and postreunion observations, each animal was observed for two 5-min focal samples, once in the first 20 min and once in the second 20 min. During the postseparation, late-separation, and prereunion periods, each separated infant was observed for four 5-min periods, and observations were alternated between cohorts. Finally, in the late-reunion period, each mother and infant was observed for one 5-min focal sample each.

Results

Preseparation

Two effects of treatment were evident in the preseparation observations. One of these treatment effects had been evident during the original early infant rearing; dyadic grooming was significantly greater within VFD dyads than it was within LFD dyads (16% of samples vs. 6% of samples), $F(1, 10) = 8.44$, $p < .05$. The other treatment group difference had not been evident during the early infant rearing; the LFD infants engaged in more nondyadic passive contact than did the VFD infants (25% of samples vs. 6% of samples), $F(1, 10) = 6.22$, $p < .05$. There were no other significant main effects of treatment group in the preseparation observations.

Infant Response to Separation

The only main effect of rearing condition during separation was for eating behavior. LFD infants ate about half as often as did the VFD infants (percentage of 30-s samples in which eating was scored, *M*s = 8% for LFD vs. 15% for VFD), $F(1, 10) = 7.20$, $p < .05$. It is unlikely that this result reflects a group difference in hunger inasmuch as both groups had had easy access to unlimited food for a minimum of 2 weeks by the time the separations were initiated. None of the remaining measures showed either significant main effects for rearing conditions or significant interactions between rearing condition and any of the other dimensions. As a consequence, the group scores were combined for all other analyses of the effects of separation.

During dyadic separations, two major patterns of infant distress, primarily in the form of coo vocalizations, were evident. First, distress was highest during the observation period immediately after separation (Table 1). Second, total infant distress dropped abruptly after the first replication (percentage of 10-s intervals in which infant distress occurred, *M*s = 22.3% for first replication, 6.7% for second replication, and 7.3% for third replication), $F(2, 20) = 38.19$, $p < .001$. As evident in Figure 1, however, the effects of repeated separations were not uniform across observation periods; unlike the response in the postseparation period and to some extent that in the late-separation period, in the prereunion period infant distress was not markedly reduced after the first replication, Replication × Observation Period, $F(4, 40) = 30.11$, $p < .001$.

There were strong positive correlations between the postseparation distress scores in the first and second separations (Pearson $r = .71$, $p < .01$) and between the late-separation distress scores of these two separations ($r = .88$, $p < .001$), whereas distress scores of the third separation appeared unrelated to those observed in the first two separations. Interestingly, the rank–order correlation for distress on the first postseparation was highly correlated with postseparation scores after the third separation (Spearman $\rho = .70$, $p < .05$); this result offers further indication of the stability of individual differences in relative levels of distress after separation. It is worth noting, as well, that levels of contact between members of the dyad in the preseparation period were moderately predictive of postseparation distress scores on the first separation ($r = .59$, $p < .05$) but not on subsequent separations. Similarly, measures of the proportions of approaches to contact and leaves from contact made by the infants before separation were unrelated to separation distress scores.

The reduction in infant distress across observation periods after separation was paralleled by the finding that exploration of inanimate objects by the infants more than doubled in the observation periods subsequent to the period immediately after separation (Table 1). However, play behavior,

Table 1
Analysis of Infant Behaviors During Separation

Behavior	Mean score per 5 min			
	Postseparation	Late-separation	Prereunion	$F(2, 20)$
Distress[a]	26	4	6	39.87***
Affiliative contact[a]	25	13	22	5.18*
Play bouts[b]	1.3	2.6	0.7	7.40**
Object exploration[b]	0.7	1.7	1.6	14.72***

[a] Percentage of interval samples. [b] Frequency.
* $p < .05$. ** $p < .01$. *** $p < .001$.

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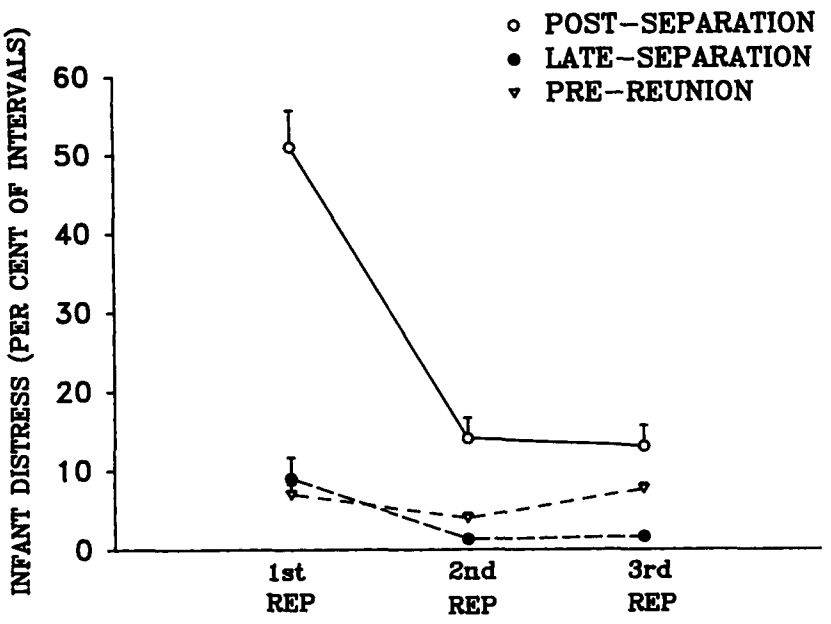


Figure 1. Infant distress as the percentage of 10-s intervals across the three replications (REP) of the postseparation, late-separation and prereunion conditions.

another measure that may also reflect infant adjustment to the separation, did not show any clear indication that infants were adjusting rapidly. Although play did rise between the postseparation and late-separation periods, it fell again in the prereunion period (Table 1).

In considering these periods after separation, it is also of interest to note that affiliative contact by the infant with its peers (passive contact and grooming) exhibited a reciprocal relationship with play behavior. Whereas play behavior was highest in the late-separation period, affiliative contact reached its highest level in the postseparation and prereunion periods. This variation in affiliative contact was significant (Table 1).

Despite the dramatically reduced infant distress after the second and third separations, there was little additional evidence that the infants were coping more effectively with the second and third separations. Although play and affiliative contact did tend to rise slightly across replications, neither behavior changed significantly. Similarly, exploration of inanimate objects remained virtually unchanged across replications.

Dyadic Reunions

As with most of the measures taken during the separations themselves, the LFD and VFD rearing groups did not differ in their behavior at reunion; the treatment groups were therefore combined for the analysis of reunion effects.

As one would anticipate, dyadic contact (passive contact and grooming) was extremely high after reunion. Whereas the mean dyadic contact score was only 43% of 30-s intervals before separation, dyadic contact was maintained during 94% of intervals after reunion, and this difference was highly significant, $F(1, 10) = 87.92$, $p < .001$. In fact, contact was scored for every interval after reunion for 5 of the 12 dyads. All 7 of the dyads that were not scored as being in contact in every interval after reunion had their lowest score in the first 5-min reunion sample; this order effect of 5-min samples in the postreunion observation period was significant (percentage of 30-s intervals in which dyadic contact was scored, *M*s = 86% for first 5 min, 98% for second 5 min, 96% for third 5 min, and 98% for fourth 5 min), $F(3, 30) = 5.23$, $p < .01$.

The low level of mother–infant contact immediately after reunion was likely due to the high level of social agitation that accompanied the return of a familiar mother to the otherwise stable group; for example, dominance–subordinance interactions in the first set of 5-min samples after reunion was four times that in the second set (frequency of dominance–subordinance interactions per 5 min, *M*s = 0.8 for first sample vs. 0.2 for second sample), $F(1, 10) = 10.76$, $p < .01$. This agitation among the adults was rapidly resolved by the onset of grooming and contact huddles; consequently, in contrast to dominance–subordinance interactions, which did not differ between preseparation and postreunion periods, affiliative contact by mothers with other group members after their return was nearly double that before their removal (percentage of 30-s intervals in which affiliative contact was scored, *M*s = 4.0% for preseparation vs. 7.5% for postreunion), $F(1, 10) = 22.37$, $p < .001$. In the agitated social milieu of a mother's return, her infant exhibited an initial peak level of departures from dyadic contact that rapidly diminished (frequency of infant departures from dyadic contact per 5-min sample, *M*s = 1.0 for first, 0.1 for second, 0.3 for third, and 0.2 for fourth), $F(1, 10) = 6.22$, $p < .05$.

By the time of the late-reunion observation (6 hr after reunion), the level of dyadic contact was considerably re-

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Table 2
Analysis of Dyadic Patterns in Late-Reunion Compared With Preseparation

Behavior	Mean frequency per 5 min		$F(1, 10)$
	Preseparation	Late-reunion	
Infant depart	3.2	2.7	4.59
Mother depart	1.4	1.0	5.44*
Infant rejection	0.2	0.4	3.37

* $p < .05$.

duced from that seen immediately after reunion. In the late-reunion period, the level of contact was 51% of 30-s intervals in contrast to the 94% of intervals in contact in the postreunion period. Furthermore, the level of dyadic contact in the late-reunion period, although slightly higher, was not significantly different from that in the preseparation period (51% vs. 43% of 30-s intervals).

Leaves from contact with the dyadic partner made by both mothers and infants were also clearly rebounding by the time of the late-reunion observations. The frequency with which infants left contact did not differ significantly between preseparation and late-reunion observations (Table 2). The frequency with which mothers broke contact in the late-reunion was approaching that of the preseparation period, but the level in late-reunion was significantly lower than that of preseparation (Table 2). It is interesting to note that infant rejections tended to be higher in late-reunion than in preseparation (Table 2), but this difference was not significant.

No significant changes across successive reunions were evident.

Novel Room Tests

As elaborated elsewhere (Andrews & Rosenblum, 1988, 1991), the novel room tests before the separation experiences were quite effective in differentiating between the LFD and VFD rearing groups. In general, it was found that the VFD infants were much less ready to use their mothers as a secure base from which to explore the novel environment than were the LFD infants. In this report we compare the novel room tests performed before the separations to novel room tests performed after all the separations were completed.

There was no evidence that the response of either group to the novel room was affected by the separation experience. In point of fact, the trends that were evident in the first novel room series continued in the second; the LFD rearing group continued to adapt to the novel room challenge in the second series, whereas the VFD rearing group continued to show little, if any, adaptation. As in the novel room series before the separations, in the second series the LFD infants continued to make more frequent excursions from the mother per 10 min than did the VFD infants (9.5 vs. 2.4 in the first series and 7.1 vs. 3.1 in the second series), overall effect for group, $F(1, 10) = 9.63$, $p < .05$. The small decrease in LFD infant departure frequency between the first and second series was inversely related to the change in separation bout length between the series. From an estimate of separation bout length (the number of intervals out of contact divided by the number of breaks in contact by mother and infant), the LFD group showed a significant increase between novel room series, whereas the VFD group exhibited a smaller, nonsignificant change (Table 3). Consequently, LFD infants were observed to be out of contact with the mother in 51% of the interval samples in the second novel room series, up from 36% in the first series; in contrast, the VFD infants were observed to be out of dyadic contact in only 21% of interval samples in the second series, slightly up from the 15% observed in the first series. The series difference, however, failed to reach statistical significance for the LFD group, as well as for the VFD group.

The findings on social play also tended to support the view that the groups differed in adaptation to the novel room. Four of the 6 LFD infants exhibited an increase in social play between the first and second series in the novel room; the difference for the group, however, failed to reach significance. Only 2 of the 6 VFD infants exhibited any increase in social play between the two novel room series; these latter increases were less than a quarter of the magnitude of the mean increase for the 4 LFD infants (0.9 bouts per 10 min for LFD vs. 0.2 bouts for VFD).

Discussion

The separation paradigm used in this study did not differentiate between two groups that had been shown by a novel environment assessment to differ in a manner consistent with a difference in attachment security. The greater effectiveness of the novel environment paradigm in relation to that of the separation paradigm in differentiating treatment groups in this study was further supported by the observation that the novel environment tests continued to differentiate the groups after the intervening sequence of dyadic separations.

Several factors may help to account for the fact that the separation paradigm used in this study did not differentiate treatment groups as the novel room tests did. With respect to separation procedures in general, it is clear that a behavioral measure taken in the absence of the mother can be, at best, a very indirect measure of attachment between infant and mother. It is also the case, in dyadic separations in nonhuman primates, that the period immediately after separation is typically dominated by distress in the infant, and depression begins to emerge in vulnerable subjects at a later time;

Table 3
Comparison of Low Foraging Demand (LFD) and Variable Foraging Demand (VFD) Groups in the Two Novel Environment Series

Group	Mean no. separation intervals per departure		$F(1, 5)$
	First series	Second series	
LFD	0.65	1.01	7.67*
VFD	0.67	0.86	0.19

* $p < .05$.

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thus the 1-day separations of the current study may have reflected elements of response to separation common to infants from a wide range of rearing conditions (Rosenblum & Plimpton, 1981). Longer separations, such as the 2-week separation used in Plimpton and Rosenblum's (1983) foraging-demand rearing study, may be necessary to allow time for more sustained differential effects due to differences in capacity for independent functioning to emerge.

More specifically with respect to the current study, it has generally been found that a separated bonnet macaque infant that is left in a familiar social milieu is quite resistant to the adverse effects of maternal separation (e.g., Reite, Kaemingk, & Boccia, 1989; Rosenblum & Kaufman, 1968); response to dyadic separation, particularly those of relatively brief duration, therefore may be a particularly unreliable measure of attachment security in this species.

In addition to the failure of treatment conditions to predict the distress responses of infants to maternal separation, dyadic interactions before separation also failed to predict the separation responses of the infants. This contrasts with the findings of Hinde and Spencer-Booth (1971), who found that in rhesus monkeys, infant separation distress was related to preseparation rejections and to the infant's role in maintaining contact with its mother.

Although Ainsworth et al. (1971) found that reunion behaviors by the human infant provided a sensitive measure of attachment security, reunion behaviors did not differentiate treatment groups in this study. Both groups exhibited a very high initial level of dyadic contact, which was approximately halved 6 hr later. Unlike their human counterparts, nonhuman primate mothers are made fearful by the imposed separation procedures and cannot be instructed how to act toward their infants on reunion. The strong protective maternal response that is consequently elicited in monkeys by the experimental procedures may therefore obscure any subtle differences in the response of infants toward their mothers at reunion. As a result, it is unlikely that reunion patterns in nonhuman primates, at least in the short term, can be reliably used to distinguish among infant responses at reunion. A paradigm, such as the novel room test, that challenges the dyad, but that does not present a challenge of such magnitude that subtle differences are masked, may therefore be viewed as a highly useful test for assessing the status of the dyadic relationship (see also Suomi, 1987).

Although the separations were not effective in differentiating the groups, significant effects of the separations were obtained for the combined groups. Both groups showed marked distress immediately after separation, with a rather rapid drop in the hours after separation. Another salient finding was the significant decrease in distress behaviors across the three separations experiences. Rather rapid reductions in infant distress patterns across repeated separations has not been consistently reported in the primate literature. Although some investigators who used different species have found results similar to those of this study (e.g., Coe et al., 1985; Suomi, Mineka, & DeLizio, 1983), others have found that protest and agitation did not decrease, or actually increased, across repeated separations in both mother–infant dyads (Jordan et al., 1984) and peers (Mineka, Suomi, & DeLizio, 1981; Mitchell, Harlow, Griffin, & Moller, 1967; Suomi et al., 1970). Factors affecting the outcome of repeated separations from an attachment figure are clearly in need of further investigation (e.g., Jordan et al., 1984).

In conclusion, the brief (i.e., less than 1 day) dyadic separations used in this study did not appear to be a robust or powerful method of assessing developmental outcome in bonnet macaque infants reared under different conditions of environmental demand for their mothers. Brief (i.e., approximately 1 hr) novel room assessments, in contrast, appear to offer a sensitive and reliable means of evaluating developmental outcome without imposing a condition, such as dyadic separation of extended duration, which itself may have long-term consequences for the development of the infant. The novel room assessments pointed to an inverse relationship between infant attachment security and the amount of demand placed on the mother by feeding contingencies during early infant rearing.

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